ECON 101 Discussion Section

Week 6. Sampling Distributions

- **Population**: The total set of subjects in which we are interested.
- **Sample**: Subset of the population for whom we have data.
- A **population parameter** is a numerical summary of the population.
- A **sample statistic** is a numerical summary of a sample taken from the population.
- The **sampling distribution** of a statistic is the probability distribution that specifies probabilities for the possible values the statistic can take.

Numerical summary	Population	Sample (size of n)
Mean	μ	$\bar{X}$
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

*A sample should be randomly chosen!

If we repeat the random sampling and calculate a sample statistic each time, we are very likely to have different values! i.e., We get a sampling distribution of the sample statistic. And, if the sample statistic is normally distributed, we can use empirical rules and z-scores to calculate a probability!

Sample Mean $\bar{X}$

The **sampling distribution** of the sample mean ($\bar{X}$) is

(1) normally distributed ***if*** *a population distribution (X) is normally distributed*, OR

(2) approximately normally distributed ***if*** a population distribution is *not* a normal distribution, but *the sample size is large enough*! ($n \geq 30$) "Central Limit Theorem (CLT)"

When the distribution of sample mean $\bar{X}$ is normally distributed,

the **mean** of $\bar{X}$: μ

the **standard deviation** of $\bar{X}$: $\frac{\sigma}{\sqrt{n}}$

Sample Proportion $\hat{p}$

p: proportion of times that x=1 in the entire population.

$\hat{p}$: proportion of times that x=1 in a sample.

If *np and n(1-p) are at least 15*, the **sampling distribution** of a sample proportion is approximately a normal distribution.

For a random sample of size n from a population with proportion p,

the **mean of $\hat{p}$** : p

the **standard deviation of $\hat{p}$** : $\sqrt{\frac{p(1-p)}{n}}$

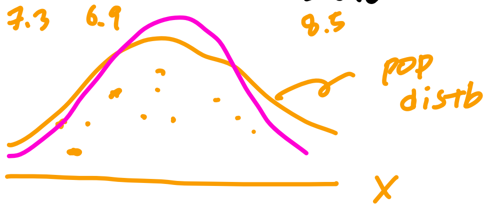
22,646 UCR undergrads

Q1 X = hrs of sleep

POP $\xrightarrow[\text{(random)}]{n=100}$ SAMPLE — SAMPLE MEAN

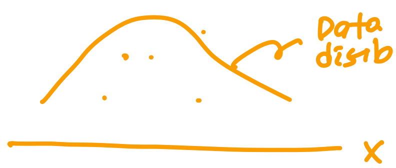
$X_1 \quad X_2 \quad \cdots \quad X_{22646}$

7.3 6.9 8.5

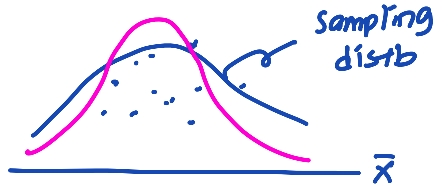


$X_2 \quad X_{91} \quad \cdots \quad X_{22600} \rightarrow \bar{x}$

6.9 4 7



$\bar{x}^1 \quad \bar{x}^2 \quad \cdots \quad \bar{x}^{70}$



mean $\mu = \dfrac{X_1 + X_2 + \cdots + X_{22646}}{N}$ ← 22646

sd $\sigma = \sqrt{\dfrac{(X_1-\mu)^2 + \cdots + (X_{22646}-\mu)^2}{N}}$

pop parameters

$\bar{x} = \dfrac{X_2 + \cdots + X_{22600}}{n}$ ← 100

$s = \sqrt{\dfrac{(X_2-\bar{x})^2 + \cdots + (X_{22600}-\bar{x})^2}{n-1}}$

Sample statistics

mean of $\bar{x} = \mu$

sd of $\bar{x} = \dfrac{\sigma}{\sqrt{n}}$

if $\underbrace{\bar{x} \text{ is normally distributed}}$

① pop $\sim N$

<u>OR</u>

② $n \geq 30$

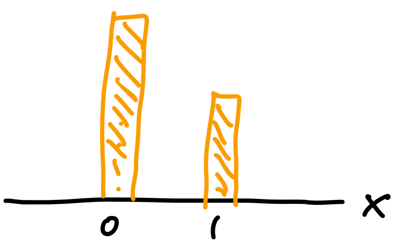
Q2 Morning person ?

$X = \begin{cases} 1 & \text{(yes)} \\ 0 & \text{(NO)} \end{cases}$

POP

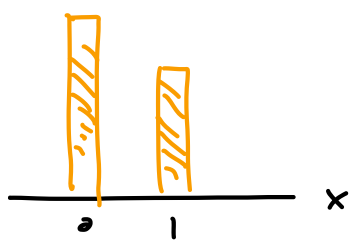
$X_1 \quad X_2 \quad \cdots \quad X_{22646}$

0 1 1



SAMPLE

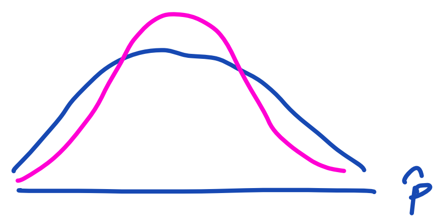
$X_2 \quad X_{91} \quad \cdots \quad X_{22600} \rightarrow \hat{p}$



SAMPLE PROPORTION

$\hat{p}^1 \quad \hat{p}^2 \quad \cdots \quad \hat{p}^{70}$

0.47 0.55 0.6



PROPORTION $p = \dfrac{X}{N}$ ← # yes

$\hat{p} = \dfrac{X}{N}$

mean of $\hat{p} = p$

sd of $\hat{p} = \sqrt{\dfrac{p(1-p)}{n}}$

if $\underbrace{\hat{p} \text{ is normally distributed}}$

① $np \geq 15$ <u>AND</u> ② $n(1-p) \geq 15$

ECON 101 Discussion Section

Week 7. Statistical Inference: Confidence Intervals

Point estimate: A single number that is our "best guess" for the parameter.

Interval estimate: An interval of numbers that is believed to contain the actual value of the parameter.

Confidence Interval: **Point estimate** $\pm$ **Margin of Error**

Margin of Error: Width of an interval measured by s.d. of the sampling distribution.

Population Proportion (p)	
A $\mathbf{100(1-\alpha)\%}$ **CI** for the population proportion (p)	$= \hat{p} \pm z_{\frac{\alpha}{2}} \boldsymbol{(se)}$
A 95% CI for the population proportion (p)	$= \hat{p} \pm \mathbf{1.96}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
A 99% CI for the population proportion (p)	$= \hat{p} \pm \mathbf{2.58}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
	$n\hat{p} \geq 15$ and $n(1-\hat{p}) \geq 15$ conditions should hold!

Population Mean	
A $\mathbf{100(1-\alpha)\%}$ **CI** for the population mean (μ)	$= \bar{x} \pm t_{\frac{\alpha}{2},df} \boldsymbol{(se)}$, where $df = n - 1$
A 95% CI for the population mean (μ)	$= \bar{x} \pm \boldsymbol{t_{0.025,df}} \frac{s}{\sqrt{n}}$
A 99% CI for the population mean (μ)	$= \bar{x} \pm \boldsymbol{t_{0.005,df}} \frac{s}{\sqrt{n}}$

ECON 101 Discussion Section

Week 8. Statistical Inference: Significance Tests

A **CI** estimates a range of plausible values for an unknown population parameter.
A **significance test** checks the plausibility of statements (i.e., hypotheses) about a population parameter.

A significance test for a…

<table>
<tr><th>Population Proportion $\boldsymbol{p}$</th><th>Population Mean $\boldsymbol{\mu}$</th></tr>
<tr><td colspan="2">1. State $\boldsymbol{H_0}$ and $\boldsymbol{H_a}$.</td></tr>
<tr><td>$H_0: p = p_0$
$H_a: p \neq p_0$ (for a two-tailed test)</td><td>$H_0: \mu = \mu_0$
$H_a: \mu \neq \mu_0$ (for a two-tailed test)</td></tr>
<tr><td colspan="2">2. Choose a significance level $\boldsymbol{\alpha}$ (usually 0.05)</td></tr>
<tr><td colspan="2">3. Calculate a test statistic (z-score or t-score)</td></tr>
<tr><td>$z = \frac{\hat{p}-p_0}{sd(p_0)}$
, where $sd(p_0) = \sqrt{\frac{p_0(1-p_0)}{n}}$</td><td>If the <u>population s.d</u>. σ is known,
$z = \frac{\bar{x}-\mu_0}{sd(\mu_0)}$, where $sd(\mu_0) = \frac{\sigma}{\sqrt{n}}$
If σ is unknown and, we use <u>sample s.d</u>. s,
$t = \frac{\bar{x}-\mu_0}{se(\mu_0)}$, where $se(\mu_0) = \frac{s}{\sqrt{n}}$</td></tr>
<tr><td colspan="2">4. Find a p-value (for two-sided tests)</td></tr>
<tr><td>p-value = $P(|z| \geq \textit{the test statistic } z)$</td><td>If the test statistic is z,
p-value = $P(|z| \geq \textit{the test statistic } z)$

If the test statistic is t,
p-value = $P(|t| \geq \textit{the test statistic } t)$
using t-distribution with $df = n - 1$</td></tr>
<tr><td colspan="2">5. Make a conclusion
Reject H_0 if p-value $\leq \boldsymbol{\alpha}$</td></tr>
</table>

$\boldsymbol{H_0}$ is a statement that a population parameter takes **a particular value**.

$\boldsymbol{H_a}$ is a statement that a population parameter falls in some **alternative range of values.**

A significance level ($\boldsymbol{\alpha}$) is **the cutoff point** that we reject H_0. We reject H_0 if p-value $\leq \alpha$.

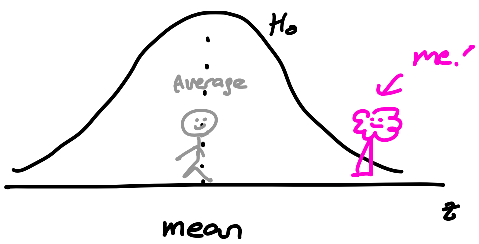
A test statistic describes how far a point estimate falls from the parameter value given in the null hypothesis. i.e., It is the distance between a sample statistic and a parameter measured in s.d. The larger the test statistic (in magnitude), the more unlikely your sample statistic value came from the null distribution.

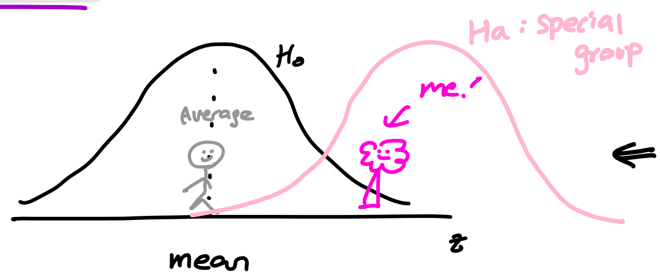
A p-value measures how unusual the data would be if H_0 were true. i.e., It is the probability that a test statistic equals the observed value or even more extreme one. The smaller the p-value, the stronger the evidence is against H_0.

H_0: You're just like the other people.

H_a: I'm special! I don't belong to a reg (normal) group.

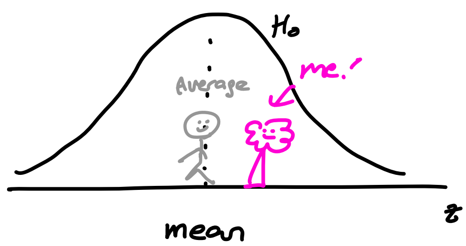
CASE 1



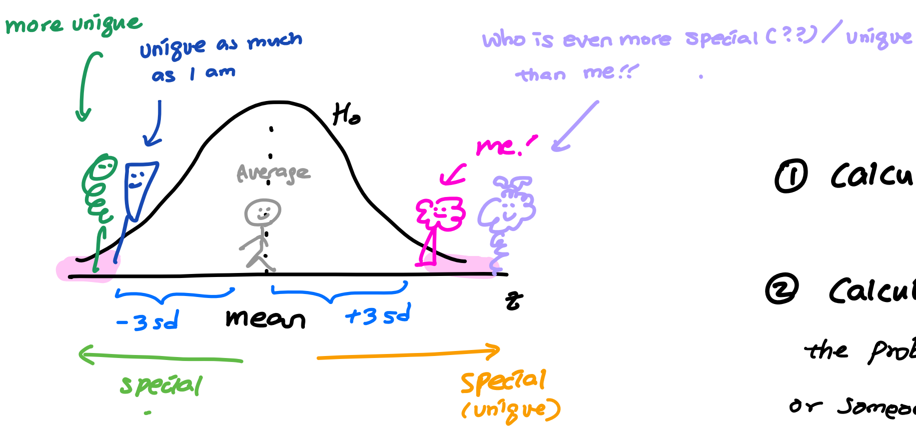


⇐ We reject H_0.

CASE 2



⇐ We cannot reject H_0.



① Calculate the test statistic.

② Calculate the p-value

the prob. that you find someone like me, or someone who is even more special / weird than me in the normal group

ECON 101 Discussion Section

Week 9. Regression Analysis

Correlation Coefficient

- Population $\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$
- Sample $r = \frac{cov(X,Y)}{s_x s_y}, where\ cov(X,Y) = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{n-1}$

  If we rearrange, it can be also written as the following:

  $$r = \frac{1}{n-1}\sum\left(\frac{x_i-\bar{x}}{s_x}\right)\left(\frac{y_i-\bar{y}}{s_y}\right) = \frac{1}{n-1}\sum(z_{xi} \times z_{yi})$$

  Don't panic. Covariance is a general form of a variance: $s_x^2 = \frac{\sum(x_i-\bar{x})^2}{n-1}$

- Correlation coefficient is between -1 and 1.
- Two variables have strong linear association if the absolute value of r is greater than 0.5.

Simple Regression Model $E[y|x] = \beta_0 + \beta_1 x$

- Interpretation:
  - $E[y|x]$: Expected value of y given x
  - β_0 (Intercept): The expected value of y when x=0
  - β_1 (Slope): The change in expected value of y when x increases by one unit.
- Error term: Actual – Expected value ⇒ $u_i = y_i - E[y|x]$

Most of the time, we don't know the true population parameters β_0 and β_1. Therefore, we estimate them with a sample. Population parameters: β_0 and β_1 ⇔ Point estimates: $\widehat{\beta_0}$ and $\widehat{\beta_1}$

We use **Ordinary Least Squares (OLS)** to estimate $\widehat{\beta_0}$ and $\widehat{\beta_1}$. OLS is a method that guesses the unknown parameters by minimizing the sum of squared residuals.

Residual = Actual – Predicted value ⇒ $e_i = y_i - \widehat{y_i}$, where $\widehat{y_i} = \widehat{\beta_0} + \widehat{\beta_1} x_i$

$$\text{OLS estimator: } \mathbf{min} \sum \boldsymbol{e_i^2} = \min \sum (y_i - \widehat{y_i})^2$$

If we solve this,

$$\widehat{\beta_1} = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} = \frac{cov(x,y)}{s_x^2} = r\frac{s_y}{s_x}$$

$$\widehat{\beta_0} = \bar{y} - \widehat{\beta_1}\bar{x}$$

We can check how our model fits well, i.e. how much our model explains the variation in the y values using **R-squared**. In a simple regression model, $R^2 = r^2$, where r = correlation coefficient.